

Software Requirements Specification

for

Local Tourist Day-Visit Planner and Information System

Version 1.0

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Revision History

Name	Date	Reason For Changes	Version
P.K.R. Abeysekara	18-Aug-2026	Initial SRS draft based on approved project proposal	1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for Version 1.0 of the Local Tourist Day-Visit Planner and Information System, a web-based application focused on the Baddegama area in Sri Lanka's Galle district. This document covers the complete system, comprising a public-facing tourist interface and a restricted administrator interface.

1.2 Document Conventions

IEEE Recommended SRS Template is followed for this document. Functional requirements are tagged with a unique identifier in the format REQ-<Feature Number>.<Sequence> (e.g., REQ-1.1). Unless a requirement explicitly states otherwise, all requirements in a System Feature inherit the priority stated for that feature in Section 4. The keyword “TBD” is used to indicate that the information is not final, and is tracked in Appendix C.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for the following readers:

- Module Examiners/Instructors – to assess completeness and correctness of the requirements.
- The Developer (the author, acting as sole developer) – to guide system design, implementation, and testing.
- Stakeholders (project group colleagues acting as prospective tourist users) – to verify that their input has been correctly captured.

The reader is encouraged to read the context and scope (Sections 1 and 2) and the interface expectations (Section 3) first, and then proceed directly to the detailed functional requirements (Section 4) and the non-functional requirements (Section 5). Testers and reviewers may prefer to go directly to Sections 4 and 5, referring back to Section 2 for context as needed.

1.4 Product Scope

The Local Tourist Day-Visit Planner and Information System is a web-based information and visit-planning application which allows tourists to find verified places of interest within a 25 km range of Baddegama and to create a simple one-day visit plan out of those places. This solves the problem of there being no reliable and updated source of local tourism information for the area by bringing together place details (category, description, opening hours, entrance fees, distance and practical visitor information) into a browsable, filterable and mappable interface. Tourism in Sri Lanka has kept on to grow strongly, with international arrivals reaching over two million visitors in 2025 [1]. An increasing share of both domestic and international travelers are increasingly relying on online resources to plan and navigate their trips [2]. The system is also of benefit to domestic and international tourists wishing to travel beyond the main coastal centers. The system does not offer booking, payment or live traffic features.

1.5 References

- [1] S. Das, “Sri Lanka’s Achieves Strong Fifteen Point One Percent Growth in Tourism with Over Two Million Visitors in 2025 Despite Decline in Per-Capita Spending - Travel And Tour World.” Accessed: Aug. 18, 2026. [Online]. Available: <https://www.travelandtourworld.com/news/article/sri-lankas-achieves-strong-fifteen-point-one-percent-growth-in-tourism-with-over-two-million-visitors-in-2025-despite-decline-in-per-capita-spending/>
- [2] D. Sinha, “How Mobile Apps are Transforming the Travel and Tourism Industry,” TechAhead. Accessed: Aug. 09, 2026. [Online]. Available: <https://www.techaheadcorp.com/blog/mobile-apps-transforming-travel-tourism-industry/>

2. Overall Description

2.1 Product Perspective

The Local Tourist Day-Visit Planner and Information System is a new, self-contained web application, it is not a follow-on to an existing product and does not replace any existing system. The system consists of three main logical components:

- (1) a public-facing tourist web client used to browse, filter, view details of, and plan visits to places of interest.
- (2) a restricted administrator web client used to maintain the place database
- (3) a backend REST API and database that serve both clients.

The application uses the services of a mapping application to map out place locations and, where possible, to suggest routes. Overall, the architecture used is the standard three-tier client/server architecture consisting of a React.js front-end, a Node.js/Express.js REST API, and a MongoDB database.

2.2 Product Functions

At a high level, the system will let users:

- Browse a categorized, filterable list of places of interest within 25 km of Baddegama.
- Search and filter places by category, distance, entrance fee, and opening hours.
- View detailed information for a selected place, including description, photos, opening hours, entrance fee, distance, and travel tips.
- View the location of places on an interactive map, either all places at once or only the places added to the current plan.
- Build, review, and edit a simple one-day visit plan by selecting and ordering places.
- (Administrator) Log in to a restricted dashboard and add, edit, or delete place records, including marking temporary closures or notices.

2.3 User Classes and Characteristics

User Class	Characteristics	Importance
Tourist User	Domestic or foreign visitor planning a one-day trip to the Baddegama area. Assumed to have general web literacy but no technical expertise and no prior familiarity with the system.	Primary user class; the system's core value proposition is built around this user.
Administrator	Single administrator who will ensure that all information in the places database is accurate. Presumed to have basic computer skills and has the proper authorization credentials.	Secondary but essential user class; keeps tourist-facing data accurate and current.

Table 1: User classes and characteristics

2.4 Operating Environment

The system will work as follows:

- Client: A modern desktop or mobile web browser (such as Chrome, Edge, Firefox, and Safari) that supports JavaScript and has internet access. The end user facing interface should be responsive enough for use on mobile phones as well since tourists will be using this system on the go.
- Frontend: React.js frontend application (HTML5, CSS3, JavaScript).
- Backend: Node.js platform with the Express.js framework serving a RESTful API.
- Database: MongoDB (a NoSQL document database); may be hosted locally or through a cloud database service such as MongoDB Atlas while in development.
- Hosting: A typical web hosting/cloud platform that can run a Node.js application; details to be decided later (refer Appendix C).

2.5 Design and Implementation Constraints

- The system must be implemented using the approved technology stack: React.js (frontend), Node.js with Express.js (backend/REST API), and MongoDB (database).
- The system must not include booking, payment, or hotel/transport reservation functionality
- The system must not depend on live/real-time traffic data for route optimization.
- Source code must be maintained under version control using Git, with the repository hosted on GitHub.
- The place database must include a minimum of 10 verified places of interest within 25 km of Baddegama.

2.6 User Documentation

The following user documentation will be delivered alongside the software: a short on-screen "How it works" guide or tooltip set for first-time tourist users covering search/filter, place details, map view, and plan building; and a brief Administrator user guide describing how to log in and add, edit, or delete place records. No printed manuals will be produced.

2.7 Assumptions and Dependencies

- It is assumed that a suitable third-party mapping service (Google Maps or an open-source alternative such as Leaflet/OpenStreetMap) provides free API access sufficient for development and demonstration purposes; the final choice is TBD (see Appendix C).
- It is assumed that place information (opening hours, entrance fees, photos, descriptions) can be verified and compiled by the developer within the project timeline, either from official sources or direct/telephone enquiry, since no live data feed from tourism authorities is assumed to be available.
- It is assumed that the four stakeholders consulted are reasonably representative of the target Tourist User class.
- The project depends on continued access to Moodle-based stakeholder consultation for any further requirement validation during SRS review.
- It is assumed that a single Administrator account is sufficient for this version of the system.

3. External Interface Requirements

3.1 User Interfaces

The tourist-facing interface will use a grid layout with large photo cards for browsing places, a prominent search bar, and simple filter controls (category, distance, entrance fee, opening hours), reflecting the layout preferences expressed by stakeholders during consultation. A place detail page will present photos, description, opening hours, entrance fee, distance, and travel tips in a clean, scannable format. The one-day visit plan will be presented as a step-by-step timeline showing each selected place in visit order, together with estimated travel time, visit duration, and running total time. The interface must be responsive and usable on both desktop and mobile screen sizes, since tourists are expected to use the system while travelling. The administrator interface will be visually different from the tourist-facing interface (for example, by a different login-protected route) and offer a dashboard with forms/lists for adding, editing, and deleting place records. Consistent navigation, visible loading and error states, and confirmation prompts prior to destructive actions (like deleting a place) are all standard UI conventions.



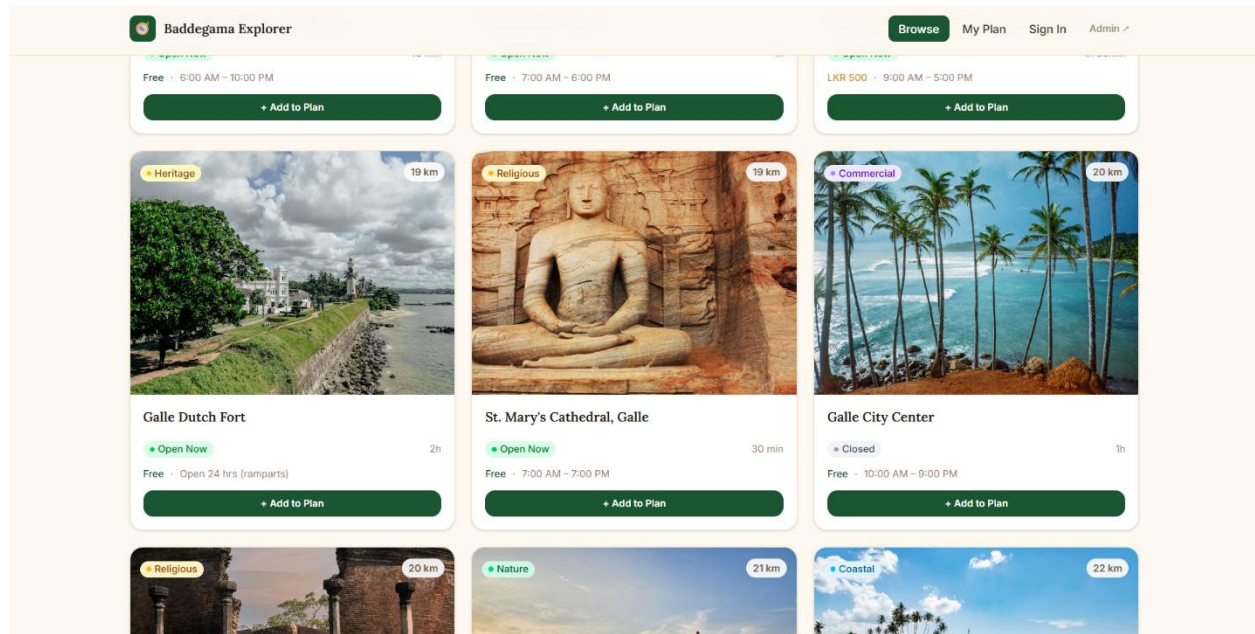
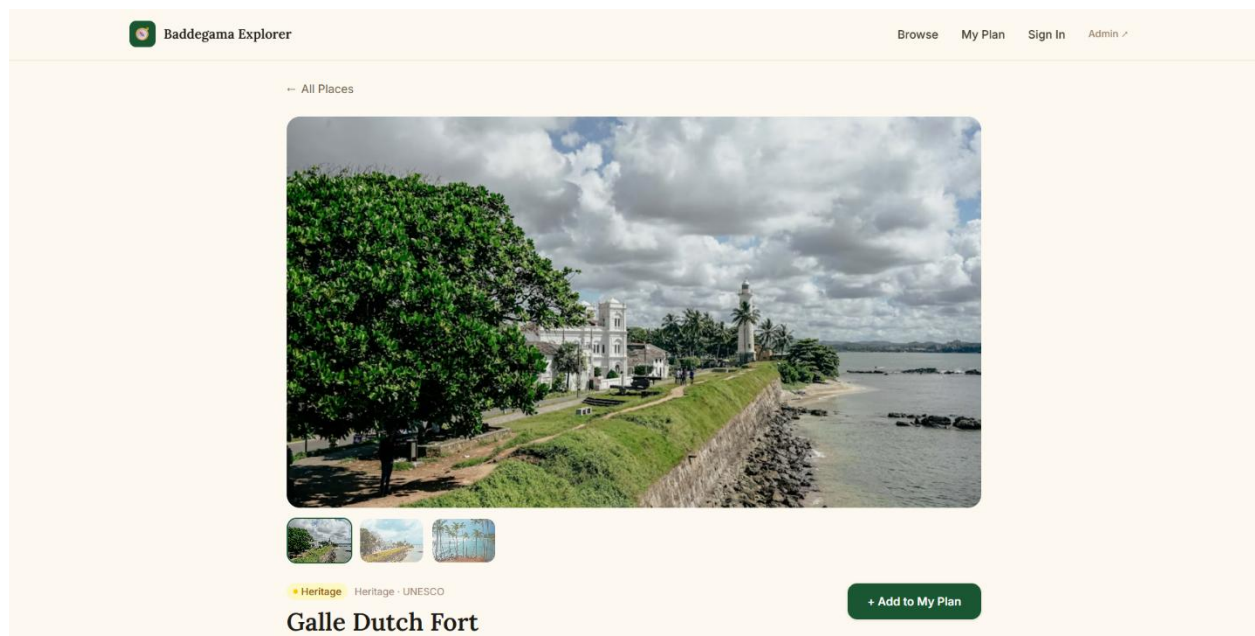


Figure 1: Tourist facing interface



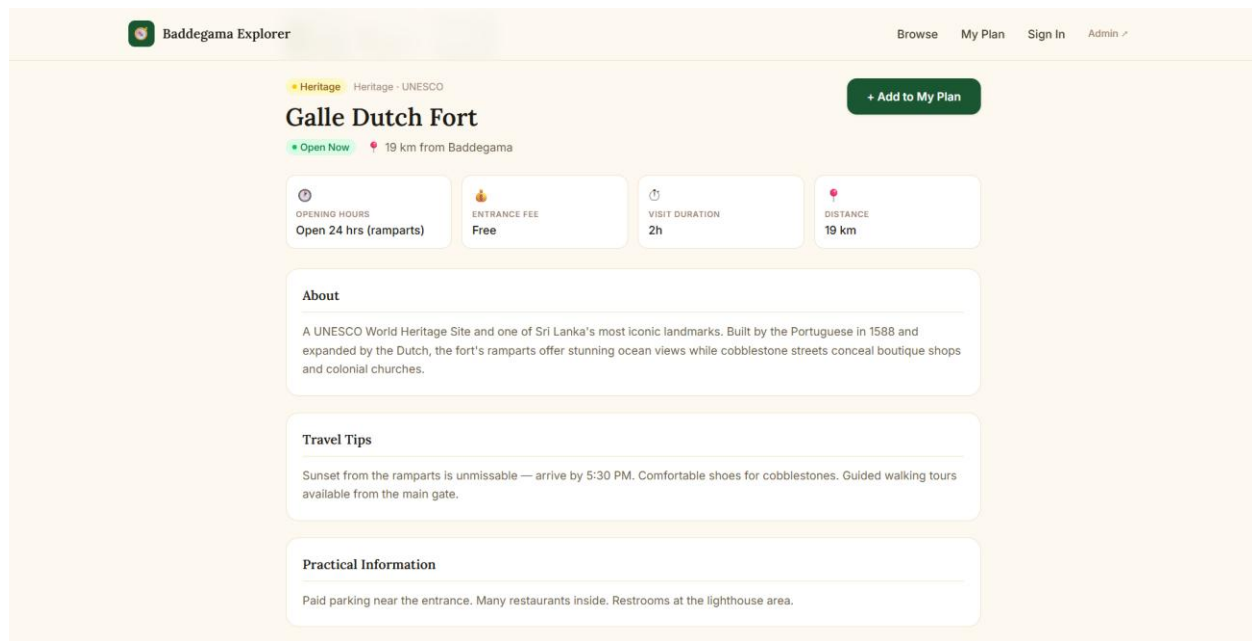


Figure 2: Place details page

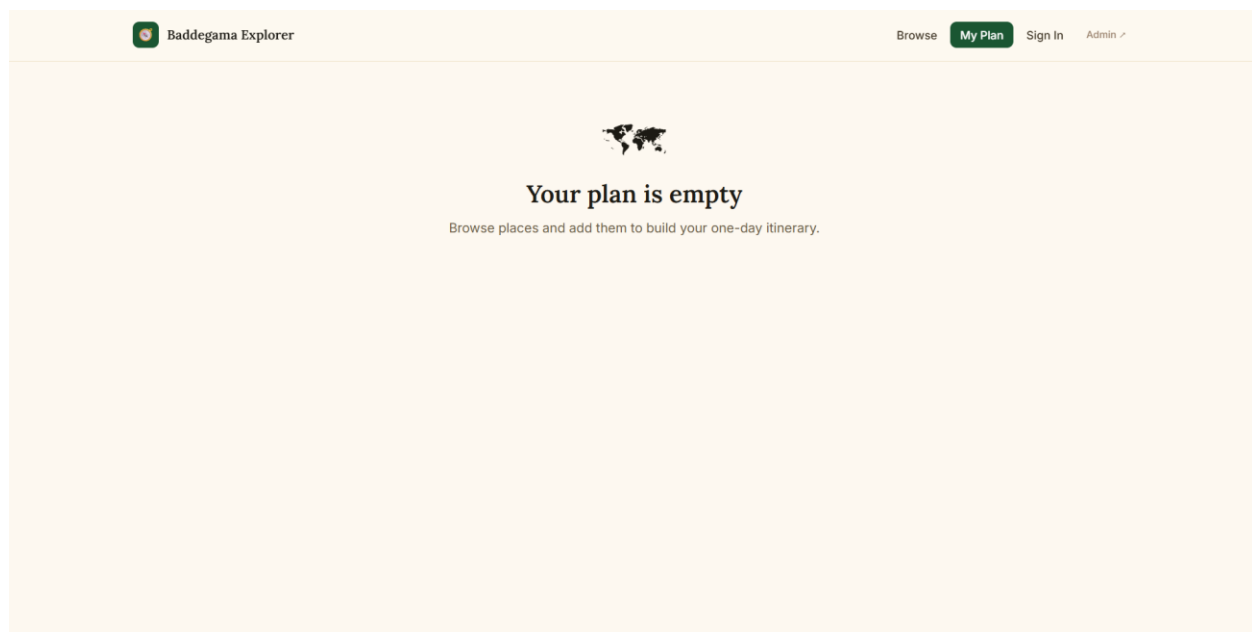
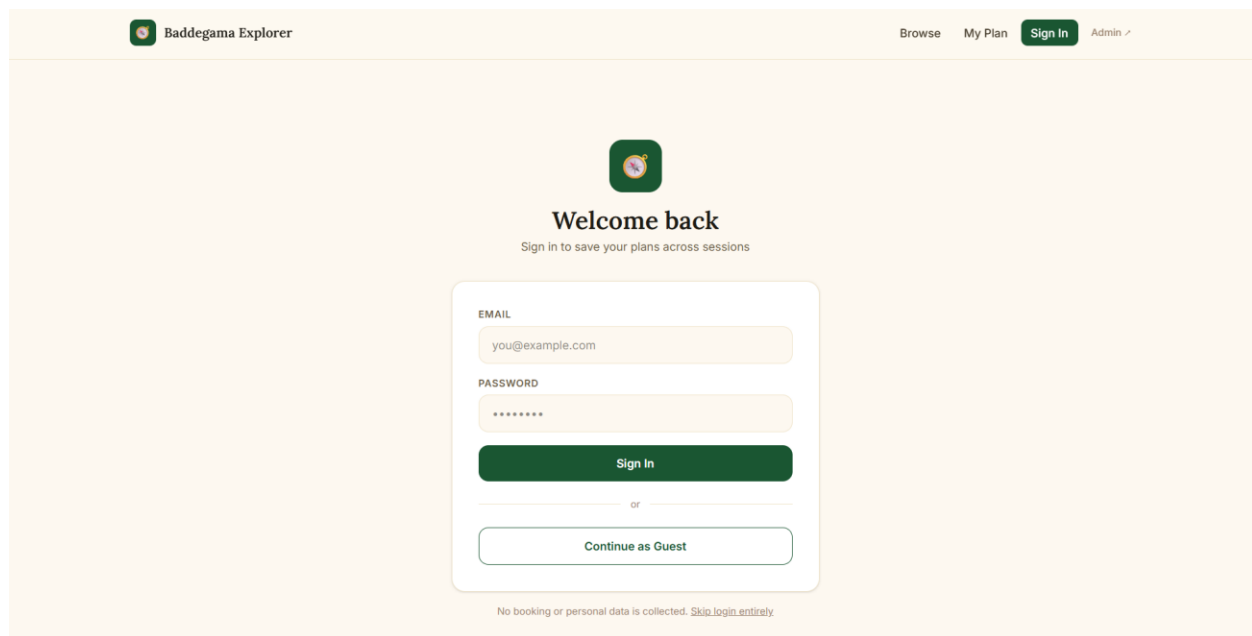


Figure 3: Visit plan interface



Baddegama Explorer

Browse My Plan Sign In Admin /



Welcome back

Sign in to save your plans across sessions

EMAIL

you@example.com

PASSWORD

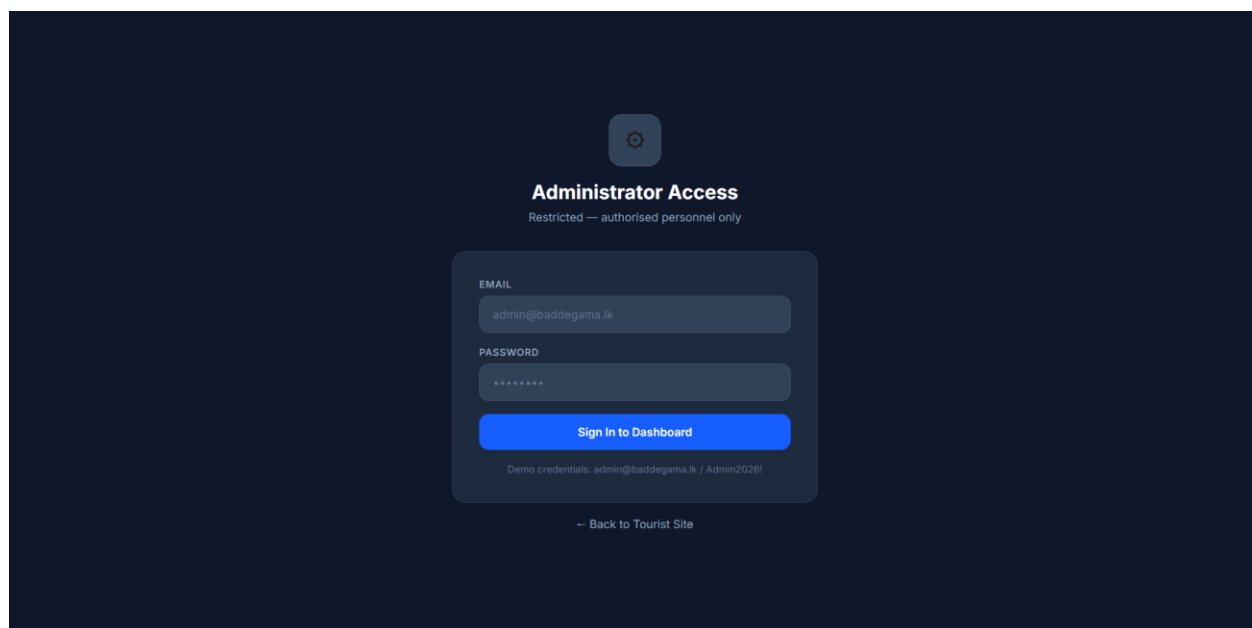
Sign In

or

Continue as Guest

No booking or personal data is collected. [Skip login entirely](#)

Figure 4: Sign In page





Administrator Access

Restricted — authorised personnel only

EMAIL

admin@baddegama.lk

PASSWORD

Sign in to Dashboard

Demo credentials: admin@baddegama.lk / Admin2026!

[← Back to Tourist Site](#)

Figure 5: Admin login

3.2 Hardware Interfaces

The system has no direct hardware interfaces. It relies on the client device's (desktop or mobile) standard input hardware (keyboard, mouse, or touchscreen) and network hardware for internet connectivity. Where the client device provides GPS/location services, the system may optionally use this to centre the map view, but this is not a required hardware dependency.

3.3 Software Interfaces

Component	Purpose / Interaction
MongoDB	Primary data store for place records, plan data, and administrator credentials. Accessed by the Node.js/Express.js backend via a database.
Node.js / Express.js REST API	Exposes JSON-based REST endpoints consumed by the React.js frontend for retrieving places, submitting plan data, and performing administrator CRUD operations. Communication uses HTTPS where deployed.
Mapping Service API	Third-party mapping/geocoding service (provider TBD, see Appendix C) used to render place locations and, where available, route information on the interactive map.
Web Browser	Renders the React.js single-page application for both the tourist and administrator interfaces.

Table 2: Software components

3.4 Communications Interfaces

JSON-formatted REST API requests and responses are used for all communication between the Node.js/Express.js backend and the React.js frontend via HTTP/HTTPS. HTTPS (TLS) must be used to protect data in transit, especially administrator login credentials, when the system is deployed to a public host. This does not require email, messaging, or other communications protocols.

4. System Features

Functional requirements are organized by system features. Each feature lists its priority, typical response sequences, and detailed functional requirements.

4.1 Categorized Place Listing and Search/Filter

4.1.1 Description and Priority

Allows tourists to browse all places of interest in a categorized, filterable grid view, and to search/filter by category, distance, entrance fee, and opening hours, as requested by stakeholders during consultation. Priority- High.

4.1.2 Stimulus/Response Sequences

- Tourist opens the home page → system displays a grid of place cards (photo, name, category, distance).
- Tourist enters a keyword in the search bar → system displays matching places in real time or on submit.
- Tourist selects one or more filters (category, distance range, entrance fee, opening hours) → system updates the displayed list to match all selected filters.
- Tourist clears filters → system restores the full list of places.

4.1.3 Functional Requirements

REQ-1.1: The system shall display all active place records in a grid of cards, each showing at minimum a photo, name, category, and approximate distance from Baddegama.

REQ-1.2: The system shall provide a search bar that filters the displayed places by matching the search text against place name and category.

REQ-1.3: The system shall allow the tourist to filter places by category such as religious, nature/recreation, heritage, coastal, shopping and dining.

REQ-1.4: The system shall allow the tourist to filter places by distance range from Baddegama.

REQ-1.5: The system shall allow the tourist to filter places by whether an entrance fee applies.

REQ-1.6: The system shall allow the tourist to filter places by opening hours availability (currently open).

REQ-1.7: If no places match the selected search criteria, the system shall display a clear "no results" message rather than an empty page.

4.2 Place Detail Information Page

4.2.1 Description and Priority

Provides a detailed information page for each place, showing the essential information tourists identified as necessary to decide whether to visit: photos, description, opening hours, entrance fee, distance, contact details (where available), and travel tips. Priority: High.

4.2.2 Response Sequences

- Tourist selects a place card from the listing or map → system navigates to the place detail page.
- Tourist views the detail page → system displays photos, description, category, opening hours, entrance fee, distance, practical information, and travel tips.
- Tourist selects "Add to plan" on the detail page → system adds the place to the current one-day plan.

4.2.3 Functional Requirements

REQ-2.1: The system shall display, for each place, its name, category, a short description, opening and closing hours, entrance fee, and approximate distance from Baddegama.

REQ-2.2: The system shall display between 3 and 5 photos for each place, where available.

REQ-2.3: The system shall display practical visitor information where available, such as parking availability, restroom availability, and nearby food options.

REQ-2.4: The system shall display travel tips for the place, such as the best time to visit, suitable clothing, or items to bring, where provided by the administrator.

REQ-2.5: The system shall provide a control on the detail page allowing the tourist to add the place directly to the current one-day visit plan.

4.3 Interactive Map View

4.3.1 Description and Priority

Displays place locations on an interactive map. Tourists may view all available places at once, or restrict the map to only the places currently selected in their one-day plan, per stakeholder feedback.

Priority- High.

4.3.2 Response Sequences

- Tourist opens the map view → system displays markers for all active places.
- Tourist toggles "Show only my plan" → system updates the map to display only places currently in the visit plan.
- Tourist selects a marker → system displays a summary popup (name, category, distance) with a link to the place detail page.

4.3.3 Functional Requirements

REQ-3.1: The system shall display the geographic location of every active place of interest as a marker on an interactive map.

REQ-3.2: The system shall allow the tourist to toggle the map between showing all places and showing only the places currently added to their one-day plan.

REQ-3.3: Selecting a marker on the map shall display, at minimum, the place name, category, and a link to its detail page.

REQ-3.4: Where supported by the mapping service, the system shall optionally display a suggested route connecting the places in the tourist's current plan in visit order.

4.4 One-Day Visit Plan Builder

4.4.1 Description and Priority

Allows a tourist to select places and organize them into a simple, ordered one-day itinerary, shown as a step-by-step timeline with estimated travel time, visit duration, and total time, as requested by stakeholders. Priority- High.

4.4.2 Stimulus/Response Sequences

- Tourist adds a place to the plan from the listing, detail page, or map → system appends the place to the current plan.
- Tourist reorders places within the plan → system updates the displayed visit order and recalculates totals.
- Tourist removes a place from the plan → system deletes it from the plan and recalculates totals.
- Tourist views the plan → system displays a timeline listing each place in order with estimated travel time between stops, estimated visit duration, opening hours, and total elapsed time for the day.

4.4.3 Functional Requirements

REQ-4.1: The system shall allow a tourist to add any active place to a one-day visit plan from the listing view, the detail page, or the map view.

REQ-4.2: The system shall allow a tourist to remove a place from the current plan.

REQ-4.3: The system shall allow a tourist to reorder the places within the current plan.

REQ-4.4: The system shall display the plan as a step-by-step timeline showing, for each place, its position in the day, opening hours, and an estimated visit duration.

REQ-4.5: The system shall display an estimated travel time between consecutive places in the plan, where distance/route data is available.

REQ-4.6: The system shall display the total estimated time required for the plan (sum of travel and visit durations).

REQ-4.7: The system shall retain the tourist's current plan for the duration of the browser session.

4.5 Administrator Place Management

4.5.1 Description and Priority

A separate interface enabling the Administrator to add, modify, and delete place records as well as update timely information, such as temporary closure and safety alerts, ensuring accuracy of data that will be used by tourists.

priority- High.

4.5.2 Stimulus/Response Sequences

- Administrator accesses the dashboard using a password → access is allowed by the system.
- Administrator adds a new place record with all necessary data → system validates and accepts it as a part of the list available for tourists.
- Administrator modifies a record → system modifies the data and changes it at the tourist's interface.
- Administrator deletes a record → system asks for confirmation and performs an action accordingly.
- Administrator makes a record temporarily closed or adds a safety alert → system shows it at the place detail page.

4.5.3 Functional Requirements

REQ-5.1: The system shall allow an authenticated Administrator to create a new place record with, at minimum, name, category, description, location coordinates, opening hours, entrance fee, and at least one photo.

REQ-5.2: The system shall allow an authenticated Administrator to edit any field of an existing place record.

REQ-5.3: The system shall allow an authenticated Administrator to delete a place record, subject to a confirmation prompt.

REQ-5.4: The system shall allow an authenticated Administrator to mark a place as temporarily closed and to add a safety or event notice, both of which shall be visible to tourists on the place detail page.

REQ-5.5: The system shall validate required fields (name, category, location) before saving a place record, and shall display a clear error message if validation fails.

REQ-5.6: The system must obtain explicit confirmation before deleting a place record; deletion cannot be undone via the standard administrator interface.

4.6 Administrator Authentication

4.6.1 Description and Priority

Preserves the integrity of the place database by limiting access to the administrator dashboard to a single authorized Administrator account.

Priority- Medium.

4.6.2 Stimulus/Response Sequences

- Administrator navigates to the admin login page and input their credentials → system verifies credentials and grants or denies access.
- Administrator access an admin only route without a valid session → system redirects to the login page.

4.6.3 Functional Requirements

REQ-6.1: The system shall require the Administrator to authenticate with a username/email and password before accessing any place management functionality.

REQ-6.2: The system shall deny access to administrator routes for any unauthenticated request and redirect the user to the login page.

REQ-6.3: The system shall store administrator passwords using a secure, one-way hashing mechanism so passwords shall never be stored or transmitted in plain text.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The place listing page shall load and display an initial set of place cards within 3 seconds under normal broadband/mobile-data conditions.
- Search and filter operations shall update the displayed results within 1 second of the tourist changing a filter, for a database of up to a few hundred place records.
- The interactive map shall render all active place markers within 3 seconds of the map view being opened.

5.2 Safety Requirements

The system provides informational travel tips (like best time to visit, suitable clothing, safety notices for specific places) but does not control or guarantee any physical safety condition at the places listed. The system shall clearly present administrator-provided safety notices where available, and shall not present unverified information as though it were an official safety guarantee. No requirement exists for safety certification, as the system does not control physical infrastructure.

5.3 Security Requirements

- Administrator access shall require authentication (see REQ-6.1–REQ-6.3); the tourist-facing interface shall not require login, as it involves no personal data collection beyond an in-session visit plan.
- All administrator sessions shall use a secure session/token mechanism, and shall expire after a period of inactivity (exact duration TBD, see Appendix C).
- All data transmitted between client and server shall be encrypted using HTTPS/TLS when the system is deployed to a public environment.
- The system shall not collect or store any tourist personal data (names, payment details) for Version 1.0, since no booking or payment features are in scope.

5.4 Software Quality Attributes

- Usability: The tourist interface shall be usable by a first-time visitor without instructions, reflecting the simple, mobile-friendly, card-based layout requested by stakeholders.
- Availability: The system should be available for demonstration and marking throughout the project evaluation period.
- Maintainability: The codebase shall follow a clear separation between frontend (React.js), backend (Express.js REST API), and data layer (MongoDB) to simplify maintenance.
- Portability: The user interface that is designed for tourists shall function properly on the latest versions of major browsers (Chrome, Firefox, Edge, Safari) on both desktop and mobile devices.
- Reliability: Core browsing and plan-building functionality shall remain usable even if the mapping service is temporarily unavailable, with a graceful fallback message shown instead of the map.

5.5 Business Rules

- Only an authenticated Administrator may create, edit, or delete place records; Tourist Users may only read place data and manage their own session-based visit plan.
- A place record must belong to at least one recognized category (religious, nature/recreation, heritage, coastal, shopping and dining, or another category defined by the Administrator) before it can be published to tourists.
- A place marked as temporarily closed by the Administrator shall remain visible in listings (clearly labelled as closed) rather than being hidden, so tourists are aware of its existence and status.

6. Other Requirements

Internationalization: Version 1.0 of the system will be presented in English only; multi-language support (Sinhala, Tamil) is not required but may be considered as a future enhancement. Legal: No specific legal or regulatory requirements apply beyond standard data-protection good practice, since the system does not process payments or store sensitive personal data. Reuse: Where feasible, the Administrator dashboard and the tourist-facing detail components should share common UI building blocks to reduce duplication of effort within the project timeline.

Appendix A: Glossary

Term	Definition
Administrator	The authorized user responsible for maintaining place records in the system.
Place of Interest	A tourist attraction, facility, or point of interest recorded in the system's database (e.g. temple, park, museum, beach).
One-Day Visit Plan	An ordered list of places, built by a tourist, intended to be visited within a single day, together with estimated travel and visit durations.
REST API	Representational State Transfer Application Programming Interface; the interface exposed by the backend server for the frontend to retrieve and modify data.
SRS	Software Requirements Specification.
Tourist User	A domestic or international visitor using the public-facing interface of the system to discover places and plan a visit.
UI / UX	User Interface / User Experience.

Table 3: Acronyms and abbreviations

Appendix B: Analysis Models

This appendix references the preliminary list of places of interest compiled during requirements analysis which forms the basis of the initial place database content.

Place Name	Category	Distance	Description
Dutugemunu Rajamaha Viharaya	Religious / Heritage	~0.8 km	A Buddhist temple in Baddegama town of cultural and historical significance.
Serene River Park	Nature / Recreation	~8.5 km	A riverside recreational park near Poddala suited to families and casual visitors.
National Maritime Archaeology Museum	Heritage / Cultural	~18 km	A museum within the Galle Fort premises showcasing maritime artefacts and shipwreck findings.
Galle Dutch Fort	Heritage	~19 km	A historic fortification built by the Portuguese and expanded by the Dutch; the historic core of Galle city.
St. Mary's Cathedral, Galle	Religious / Heritage	~19 km	A historic Roman Catholic cathedral notable for its colonial-era architecture.
Galle City Center	Commercial	~20 km	A shopping mall offering retail, dining and entertainment for visitors extending their day trip.
Yatagala Rajamaha Viharaya	Religious / Heritage	~20 km	An ancient rock temple near Unawatuna, notable for its reclining Buddha statue and murals.
Rumassala	Nature / Heritage / Religious	~21 km	A forested headland near Unawatuna, home to the Japanese Peace Pagoda with panoramic coastal views.

Jungle Beach, Unawatuna	Coastal / Nature	~22 km	A secluded cove beach sheltered by a coral reef, popular for swimming and snorkelling.
Unawatuna Scuba Diving	Adventure / Marine	~22 km	A dive centre near Unawatuna Beach offering guided scuba and snorkelling excursions.

Table 4: Preliminary list of places of interest within 25 km of Baddegama

Evidence of stakeholder consultation (forum discussion screenshots) used to derive the functional requirements in Section 4 is available in Appendix A of the Project Proposal. A summary of stakeholder input is reproduced below for traceability.

Stakeholder	Role / Focus Area	Key Requirement Identified
D.L.A.L. Denipitiya	Prospective tourist – Navigation & Trip Planning	Map navigation, suggested travel routes, and clearly defined place categories to support one-day trip planning.
T.M.V.S. Thennakoon	Prospective tourist – Content & Information	Essential place details (photos, opening hours, entrance fee, location, contact information), an itinerary builder with reorderable stops and travel-time estimates, plus weather information and safety tips.
S.M. Perera	Prospective tourist – Search & Interface	Multi-criteria search filters (category, distance, entry fee, opening hours, activity type); a step-by-step timeline day-plan view; a grid layout with photo cards and search bar; a map restricted to selected trip places with an option to view all; regular admin updates.
E.H.M.H.M. Samarasinghe	Prospective tourist – Practical Information	Basic place details, 3–5 quality photos per place, short descriptions, practical information (parking, restrooms, food), and a simple visit plan grouping nearby attractions.

Table 5: Summary of stakeholder-identified requirements

B.1 Data flow diagram (level 1)

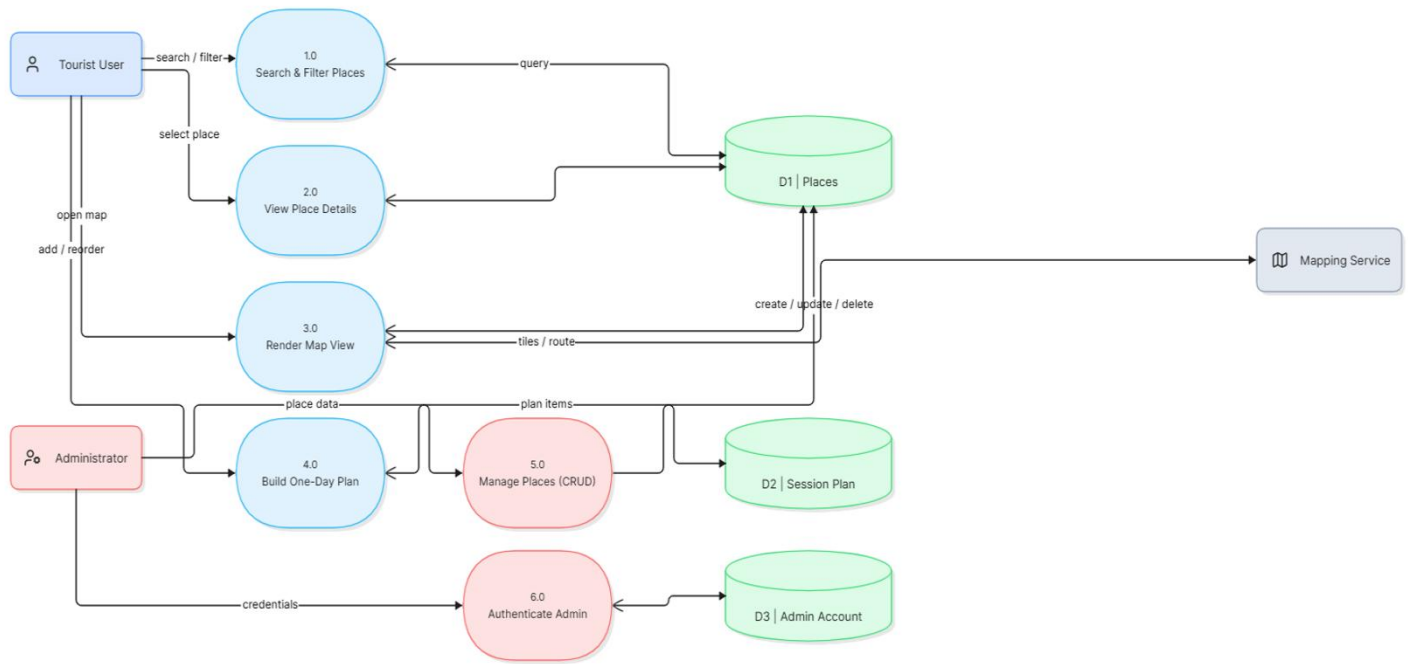


Figure 6: Data flow diagram

B.2 Entity Relationship diagram

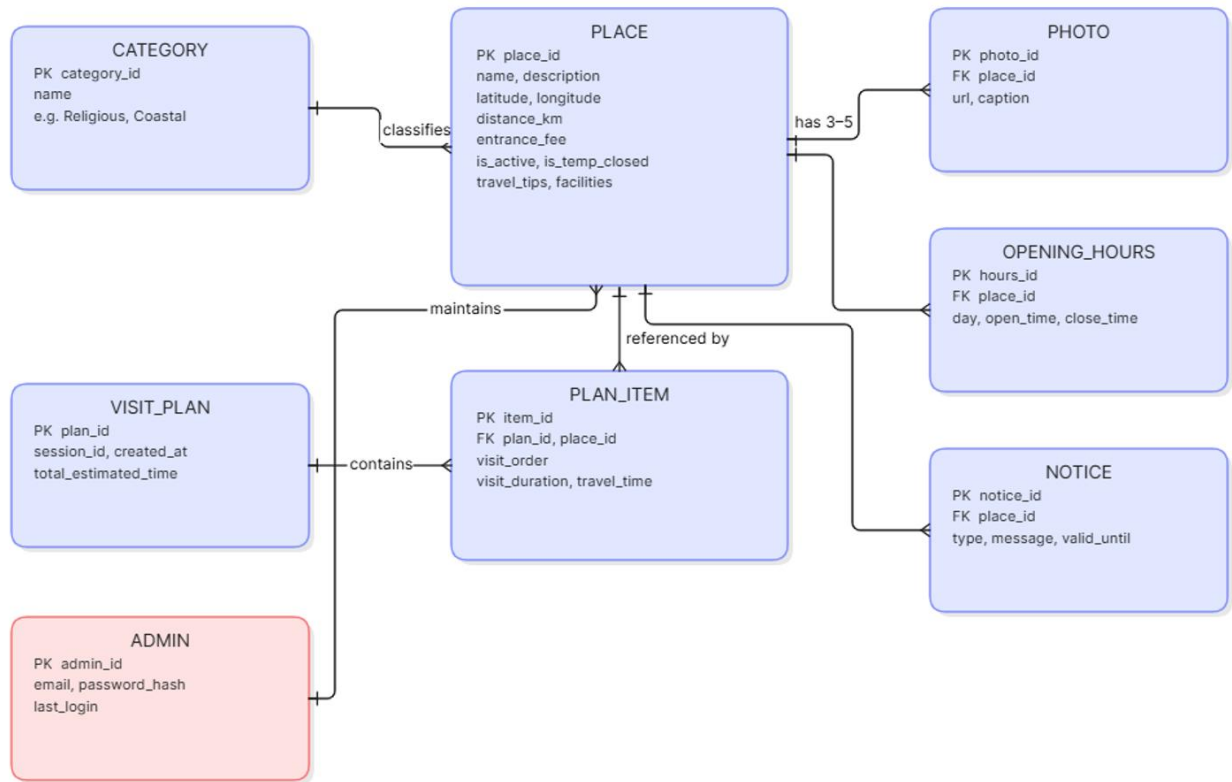


Figure 7: Entity Relationship diagram

B.3 Class diagram (MERN layer)

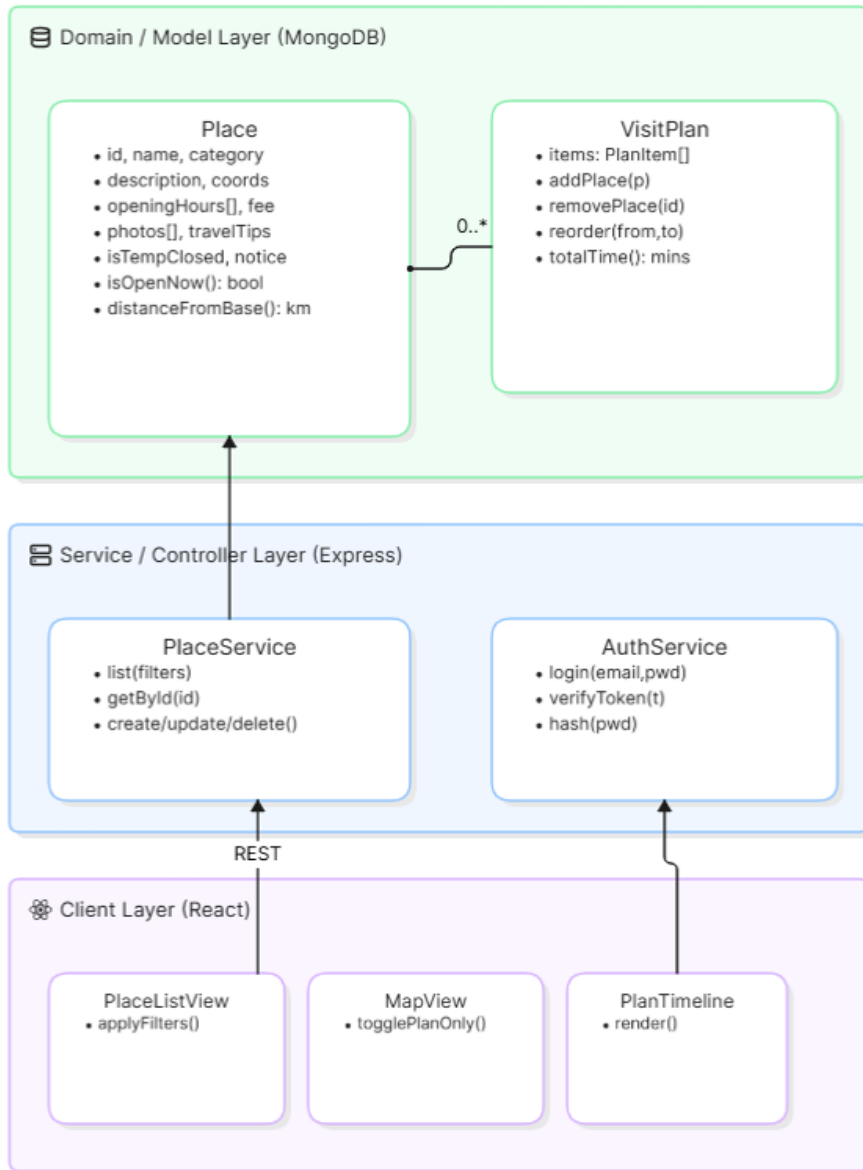


Figure 8: Class diagram (MERN layer)

B.4 State transition diagram

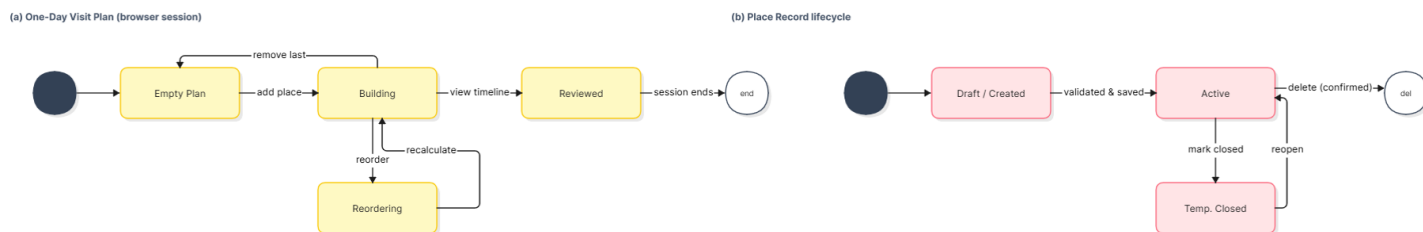


Figure 9: State transition diagram

B.5 Local Tourist Day-Visit Planner Data Model

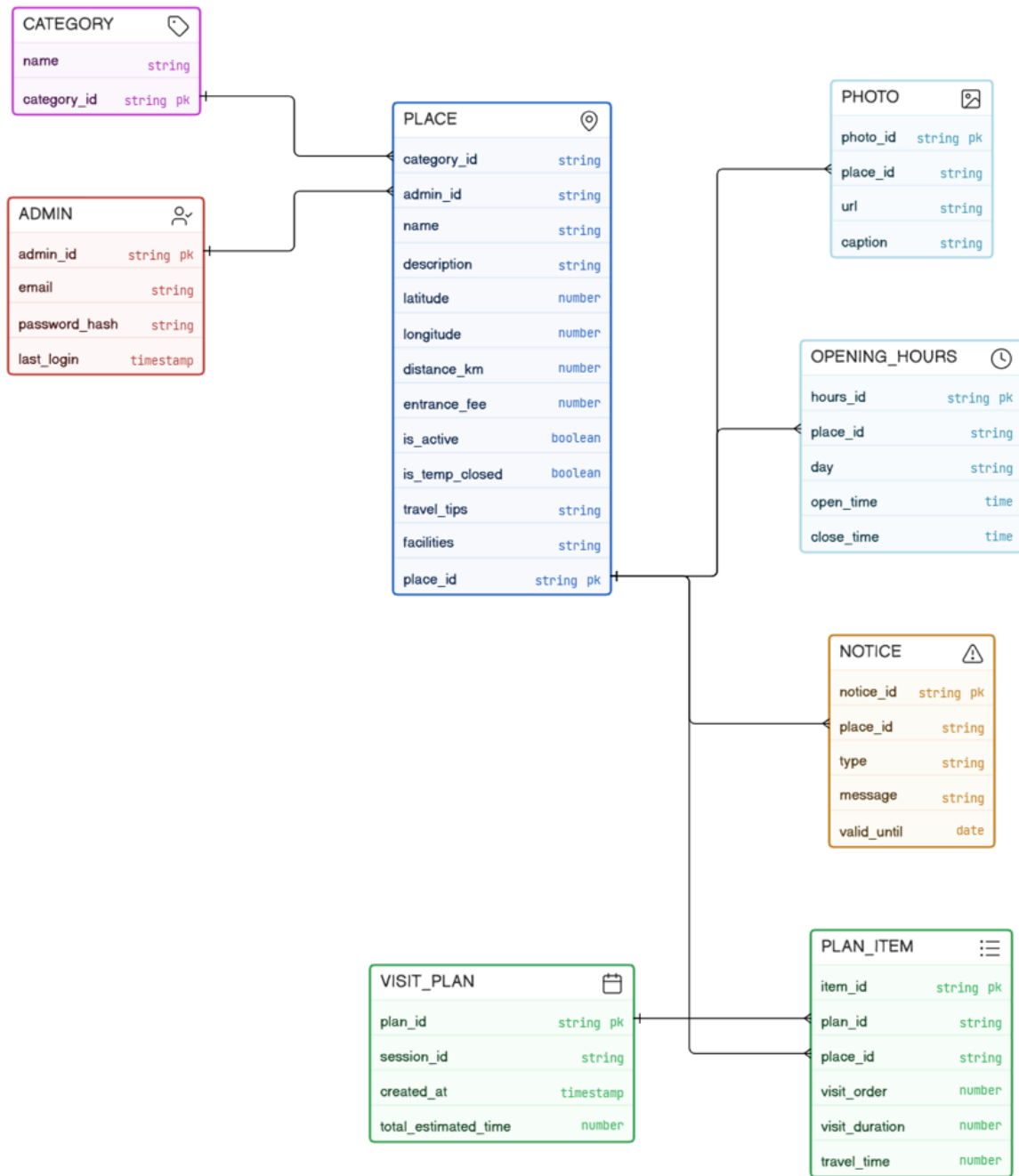


Figure 10: Visit Planner Data Model

Appendix C: To Be Determined List

- TBD-1 (Section 2.7 / 3.3): Final choice of third-party mapping/geocoding service provider (e.g. Google Maps API vs. Leaflet/OpenStreetMap) has not yet been confirmed.
- TBD-2 (Section 2.4): Final hosting/deployment platform for the production system has not yet been confirmed.
- TBD-3 (Section 5.3): Exact administrator session/token expiry duration has not yet been confirmed.
- TBD-4 (Section 4.3): Availability and cost of route/travel-time data from the chosen mapping service, which affects whether REQ-3.4 and REQ-4.5 can be fully implemented, has not yet been confirmed.